Kenji Watanabe

Machine Learning Researcher

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kenji.example.org
github.com/kenji-w-example

SKILLS

ML: PyTorch, JAX, Hugging Face Transformers, Speech recognition
Programming: Python, C++, CUDA

RESEARCH SUMMARY

PhD candidate researching efficient transformer architectures for low-resource speech recognition. Author of 3 peer-reviewed papers. Seeking research engineer roles in applied ML.

EDUCATION

PhD, Computer Science
University of Example Toronto
Sep 2021 – Present

- Thesis: Parameter-efficient adaptation of speech transformers.

M.Eng., Information Science
Sample Institute of Science
Apr 2019 – Mar 2021 | Osaka, Japan

RESEARCH EXPERIENCE

Research Intern
Litware AI Lab · Internship
May 2024 – Aug 2024 | Montreal, QC

- Implemented low-rank adapters for a 300M-parameter speech model, matching full fine-tuning accuracy with 2% of trainable parameters.
- Open-sourced the training code and evaluation scripts.

Technologies: PyTorch, JAX

Graduate Research Assistant
University of Example Toronto · Part-time
Sep 2021 – Present

- Teaching assistant for the graduate Machine Learning course (2 terms).

PUBLICATIONS

Adapter Tuning for Low-Resource Speech Recognition
Proceedings of the Example Speech Conference
2024
K. Watanabe, A. Example

Pruning Attention Heads in Streaming ASR
Example Workshop on Efficient ML
2023
K. Watanabe, B. Sample, A. Example

AWARDS

Graduate Research Fellowship
University of Example Toronto
2021